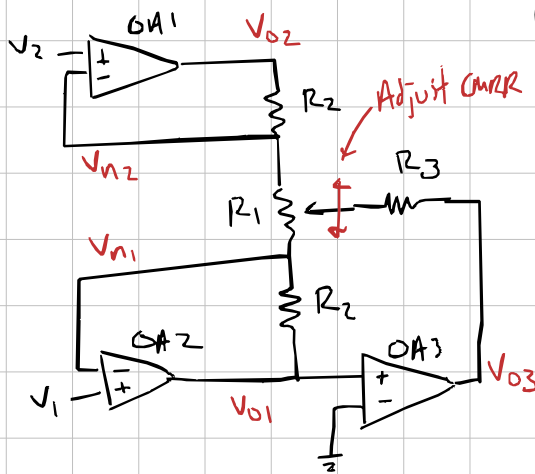


2.43



$$R_1 = 1k\Omega \text{ pot}$$

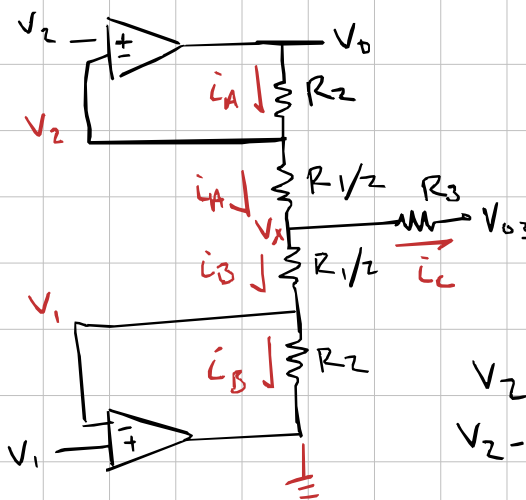
$$R_2 = 20k\Omega$$

$$R_3 = 30k\Omega$$

show that

$$V_0 = \left(1 + 2R_2/R_1\right)(V_2 - V_1)$$

Assuming circuit is stable. V_{01} must be ≈ 0 for V_{03} to not be saturated.



$$V_0 - V_2 = i_A R_2 \rightarrow i_A = \frac{V_0 - V_2}{R_2}$$

$$V_2 - V_x = i_A R_1/2 \rightarrow V_x = V_2 - i_A \frac{R_1}{2}$$

$$V_x - V_1 = i_B R_1/2 \rightarrow V_x = V_1 + i_B \frac{R_1}{2}$$

$$V_1 - 0 = i_B R_2 \rightarrow i_B = \frac{V_1}{R_2}$$

$$V_2 - i_A \frac{R_1}{2} = V_1 + i_B \frac{R_1}{2}$$

$$V_2 - \frac{V_0 - V_2}{R_2} \frac{R_1}{2} = V_1 + \frac{V_1}{R_2} \frac{R_1}{2}$$

$$-\frac{R_1}{2R_2} V_0 + \left(1 + \frac{R_1}{2R_2}\right) V_2 = \left(1 + \frac{R_1}{2R_2}\right) V_1$$

$$V_0 = \left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1) \quad \checkmark$$

$$V_1 = \frac{R_2}{R_2 + R_1/2} V_3 = \frac{1}{1 + R_1/2R_2} V_3 \rightarrow V_3 = \left(1 + \frac{R_1}{2R_2}\right) V_1 \quad \checkmark$$

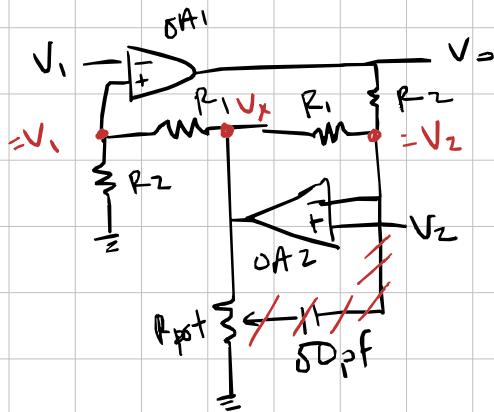
$$\frac{V_0 - V_2}{R_2} = \frac{V_2 - V_3}{R_1/2}$$

$$V_0 - V_2 = \frac{2R_2}{R_1} (V_2 - V_3)$$

$$V_0 = \left(1 + \frac{2R_2}{R_1}\right) V_2 - \frac{2R_2}{R_1} \left(1 + \frac{R_1}{2R_2}\right) V_1$$

$$V_0 = \left(1 + \frac{2R_2}{R_1}\right) (V_2 - V_1)$$

2.47]



Show $V_0 = \left(1 + \frac{R_2}{R_1}\right)(V_2 - V_1)$
at low frequency

At low frequency: capacitor acts like open circuit...
so just remove connection to OA2 V_p & potentiometer

Since R_{pot} is in parallel with connection to rest of circuit, can ignore assuming $r_o \rightarrow \infty$

V_x is driven ... find it with respect to V_1 .

$$V_1 = V_x \frac{R_2}{R_1 + R_2} \rightarrow V_x = \left(1 + \frac{R_1}{R_2}\right)V_1 \quad (1)$$

an in respect to $V_2 \hat{=} V_0$

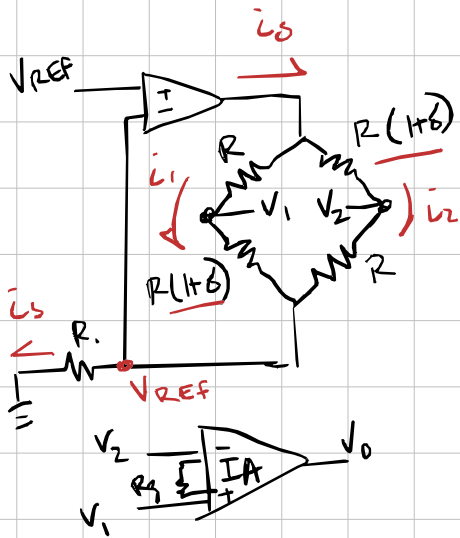
$$V_2 = (V_0 - V_x) \frac{R_1}{R_1 + R_2} + V_x \quad (2)$$

$$(1) \rightarrow (2) \quad V_2 = \left(V_0 - \left(\frac{R_1 + R_2}{R_2}\right)V_1\right) \frac{R_1}{R_1 + R_2} + \left(1 + \frac{R_1}{R_2}\right)V_1$$

$$V_2 = \frac{R_1}{R_1 + R_2} V_0 - \frac{R_1}{R_2} V_1 + \left(1 + \frac{R_1}{R_2}\right)V_1$$

$$\rightarrow V_0 = \left(1 + \frac{R_2}{R_1}\right)(V_2 - V_1) \quad \checkmark$$

2.58)



$V_{REF} = 2.5V$
want $0.1 V/^\circ C$ at output of IA.

variable resistors or 1000Ω RTDs

$$\therefore R = 1000\Omega!$$

At a maximum, target $0.1mA$ through an RTD

$$\therefore i_1 = i_2 = 0.1mA \rightarrow i_3 = 0.2mA$$

$$i_3 = V_{REF}/R_1 \text{ via virtual short}$$

$$0.2mA = 2.5V/R_1$$

$$\rightarrow R_1 = 12500\Omega$$

from Book/Lecture:

$$V_1 - V_2 = R\delta i_3/2 = 0.1\delta$$

and

$$V_0 = A(0.1\delta) \rightarrow A \text{ is gain of IA set by } R_g$$

What is δ ?

RTD has $0.00385 \Omega/\Omega/^\circ C$

$$\rightarrow \delta = 0.00385 \Omega/\Omega/^\circ C \cdot T[^\circ C]$$

$\uparrow$ Temperature in $^\circ C$

for $1^\circ C$ then, RTD will increase 0.385% , $\delta = 0.00385$

$$V_0[1^\circ C] = 0.1V/^\circ C = A(0.1)(0.00385)$$

$$\rightarrow A = 259.74 \approx 260$$

Summary: $R = 1k\Omega$, $R_1 = 12.5k\Omega$
 $A = 260 V/V$

Since A will depend on R_g & IA internal gain, cannot select R_g without knowing IA

Power consumed by resistor (assume $0^\circ C$, nominal):

$$P_{R1} = V_{REF} i_3 = 2.5V \cdot 0.2mA = 500\mu W$$

$$P_R = i_1^2 R = i_2^2 R = (0.1mA)(1k\Omega) = 10\mu W$$

$\uparrow$ all resistors in bridge

$$P_{total} = P_{R1} + 4 \cdot P_R = 540\mu W$$